

Auction Human data

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1 Human experiment data

Unified deviation metric. Consider an auction format m and an information object $\mathcal{I}$ observed by the bidder (e.g., $\mathcal{I} = v$ in private values, or $\mathcal{I} = s$ in common values). Let b denote an observed bid and let $b_m^*(\mathcal{I})$ denote the *theoretic optimum (equilibrium) bid* under format m . We define the absolute bid error

$$d_m \equiv |b - b_m^*(\mathcal{I})|.$$

To compare deviations across mechanisms on a common scale, we normalize by the *expected optimum bid level*

$$\mu_m^* \equiv \mathbb{E}[b_m^*(\mathcal{I})],$$

and define the *Scaled Mean Absolute Deviation (SMAD)* as

$$\Delta_m \equiv 100 \cdot \frac{\mathbb{E}[d_m]}{\mu_m^*} = 100 \cdot \frac{\mathbb{E}[|b - b_m^*(\mathcal{I})|]}{\mathbb{E}[b_m^*(\mathcal{I})]}.$$

The resulting metric is interpretable as a percent of the expected equilibrium bid. Given a sample $\{(b_k, \mathcal{I}_k)\}_{k=1}^n$, the plug-in estimator is

$$\widehat{\Delta}_m = 100 \cdot \frac{\frac{1}{n} \sum_{k=1}^n |b_k - b_m^*(\mathcal{I}_k)|}{\widehat{\mu}_m^*}, \quad \text{SE}(\widehat{\Delta}_m) = 100 \cdot \frac{\text{SE}(\widehat{\text{MAD}}_m)}{\widehat{\mu}_m^*},$$

where $\widehat{\text{MAD}}_m = \frac{1}{n} \sum_{k=1}^n |b_k - b_m^*(\mathcal{I}_k)|$ and $\widehat{\mu}_m^*$ is the implied mean optimum bid under the same data-generating process as the paper. We report 95% confidence intervals as $\widehat{\Delta}_m \pm 1.96 \text{SE}(\widehat{\Delta}_m)$. For presentation only, we cap plotted axes at 100; reported values are uncapped.

Common values (CV-2, welfare/profit deviation proxy). In common-value auctions, the bid-optimal benchmark $b^*(s)$ depends on the winner's-curse correction through the posterior $\mathbb{E}[V \mid s, \text{win}]$. Because many classic CV studies do not report the bid-factor $\phi = b/\mathbb{E}[V \mid s, \text{win}]$ (or sufficient information to reconstruct it), we operationalize the CV effect size using *profit deviations* from the theoretical optimum. Let π denote the realized bidder profit and π^* the theoretical predicted profit (reported by the paper). We define the CV scaled absolute profit deviation

$$\Delta_m^{\text{CV}} \equiv 100 \cdot \frac{\mathbb{E}[|\pi - \pi^*|]}{\mathbb{E}[\pi^*]}.$$

1.1 Methodology

Mapping original reported statistics into the unified metric. (Li; Breitmoser–Schweighofer–Kodritsch) AC/AC-B/2P under APV. These papers directly report a Mean Absolute Deviation (MAD) from truthful bidding, i.e., $\text{MAD}_m = \mathbb{E}[|b - v|]$ (or, in Li, a group-level statistic based on the second-highest bid and value). Because the equilibrium optimum under AC and 2P is $b^*(v) = v$, we have $d_m = |b - v|$ and thus

$$\Delta_m = 100 \cdot \frac{\text{MAD}_m}{\mu_m^*}, \quad \mu_m^* = \mathbb{E}[v].$$

In practice, $\mathbb{E}[v]$ is implied by the experimental value distribution reported by the paper (e.g., $\mathbb{E}[v] = 70$ for both Li and Breitmoser–Schweighofer–Kodritsch). If the paper reports $\text{SE}(\text{MAD}_m)$ (often with group means as the unit of observation), we propagate it to the scaled metric via

$$\text{SE}(\widehat{\Delta}_m) = 100 \cdot \frac{\text{SE}(\widehat{\text{MAD}}_m)}{\widehat{\mu}_m^*}, \quad \text{CI}_{95\%}(\Delta_m) = \widehat{\Delta}_m \pm 1.96 \text{SE}(\widehat{\Delta}_m).$$

(Kagel–Levin 1993) First-/Second-/Third-price IPV. Kagel–Levin report reduced-form bid functions $\hat{b}(x) = \hat{\alpha} + \hat{\beta}x$ along with overbidding/truthful bidding/ underbidding frequencies. Theory gives linear equilibrium bid functions under IPV:

$$b_{\text{FPA}}^*(x) = \beta_{\text{FPA}}^* x, \quad \beta_{\text{FPA}}^* = \frac{N-1}{N}; \quad b_{\text{SPA}}^*(x) = x; \quad b_{\text{TPA}}^*(x) = \beta_{\text{TPA}}^* x, \quad \beta_{\text{TPA}}^* = \frac{N-1}{N-2}.$$

Because only summary moments are reported, we reconstruct the bid distribution via a parametric data-generating process (DGP) consistent with Table 2: we assume $x \sim U[x_{\min}, x_{\max}]$ (with bounds from the paper), and generate bids as

$$b = \hat{\alpha} + \hat{\beta}x + \varepsilon, \quad \varepsilon \sim \mathcal{N}(\mu, \sigma^2),$$

where (μ, σ) are calibrated to match the reported frequencies (e.g., $\Pr(b > x)$ and $\Pr(b > b^*(x))$) for each auction series. Given the calibrated DGP, we estimate

$$\widehat{\text{MAD}}_m = \mathbb{E}[|b - b^*(x)|], \quad \widehat{\mu}_m^* = \mathbb{E}[b^*(x)], \quad \widehat{\Delta}_m = 100 \cdot \frac{\widehat{\text{MAD}}_m}{\widehat{\mu}_m^*}.$$

We implement this reconstruction using Monte Carlo simulation, repeating the procedure 100 times and reporting the average $\widehat{\Delta}_m$ across runs. Standard errors and 95% confidence intervals are computed from the across-run variability of $\widehat{\Delta}_m$.

(Gneezy–Smorodinsky 2006) All-pay. In the complete-information all-pay game with prize value V and N symmetric bidders, the Nash equilibrium is mixed; a robust first-moment implication is $\mathbb{E}[b^*] = V/N$. Using reported summary statistics (or reconstructed values from figures when necessary), we compute the mean absolute deviation from the mean equilibrium benchmark $b^* \equiv V/N$:

$$\text{MAD}_{\text{AP}} = \mathbb{E}[|b - b^*|], \quad \mu_{\text{AP}}^* = \mathbb{E}[b^*] = V/N, \quad \Delta_{\text{AP}} = 100 \cdot \frac{\text{MAD}_{\text{AP}}}{\mu_{\text{AP}}^*}.$$

(Kagel–Levin 1986; Levin–Kagel–Richard 1996 and related CV papers) First-price CV and English (second-price proxy). Under CV, bidding deviations are commonly diagnosed via winner’s-curse consequences. When the bid-factor $\phi = b/\mathbb{E}[V \mid s, \text{win}]$ is not reported, we use profit-based deviations. Let $\bar{\pi}$ and $\text{SE}(\bar{\pi})$ denote the reported mean profit and its standard error, and let $\bar{\pi}^*$ and $\text{SE}(\bar{\pi}^*)$ denote the theoretical predicted mean profit and its reported standard error (when available). We map these to the scaled absolute profit deviation proxy

$$\widetilde{\Delta}_m^{\text{CV}} = 100 \cdot \frac{|\bar{\pi} - \bar{\pi}^*|}{\bar{\pi}^*}.$$

To obtain confidence intervals, we treat $(\bar{\pi}, \bar{\pi}^*)$ as noisy summaries and propagate their uncertainty via Monte Carlo: we draw $\pi^{(r)} \sim \mathcal{N}(\bar{\pi}, \text{SE}(\bar{\pi})^2)$ and $\pi^{*(r)} \sim \mathcal{N}(\bar{\pi}^*, \text{SE}(\bar{\pi}^*)^2)$, compute $\widetilde{\Delta}_m^{\text{CV},(r)} = 100 \cdot |\pi^{(r)} - \pi^{*(r)}|/\pi^{*(r)}$, repeat for $r = 1, \dots, 100$, and report the mean and the 2.5–97.5 percentiles as the point estimate and 95% confidence interval. We also pooled the data of all the small/ medium/ large signal in the paper.

References